

## **Experiment 2: Client–Server Communication Between Two Virtual Machines**

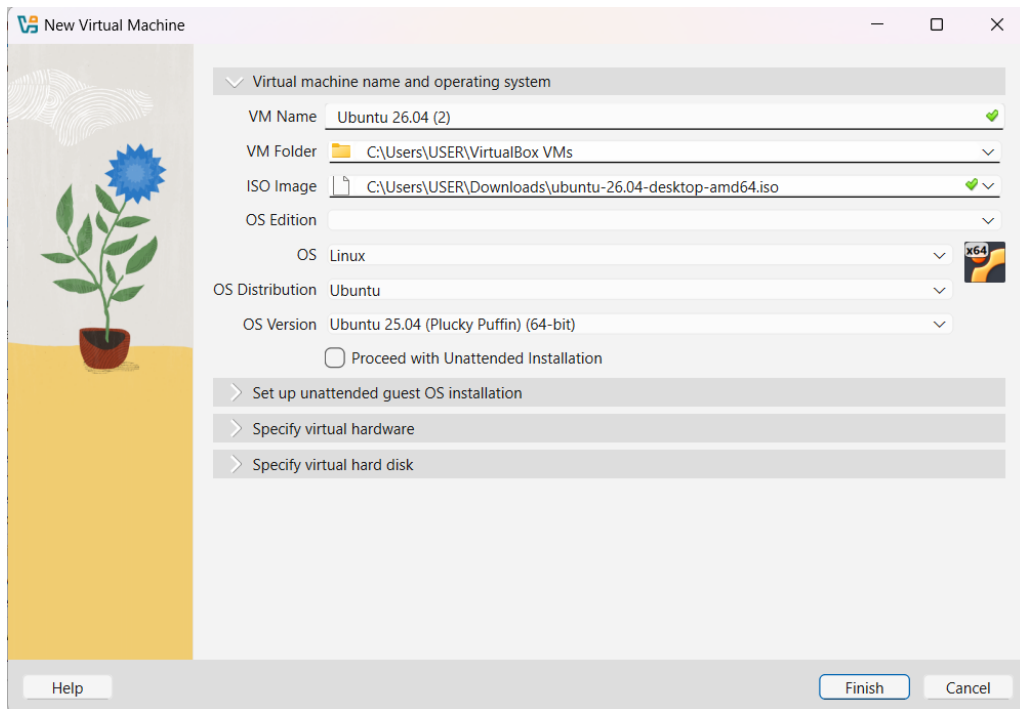
Configure two virtual machine instances in VirtualBox to establish client–server communication and implement a basic TCP-based chat application between the two virtual machines.

### **Objective**

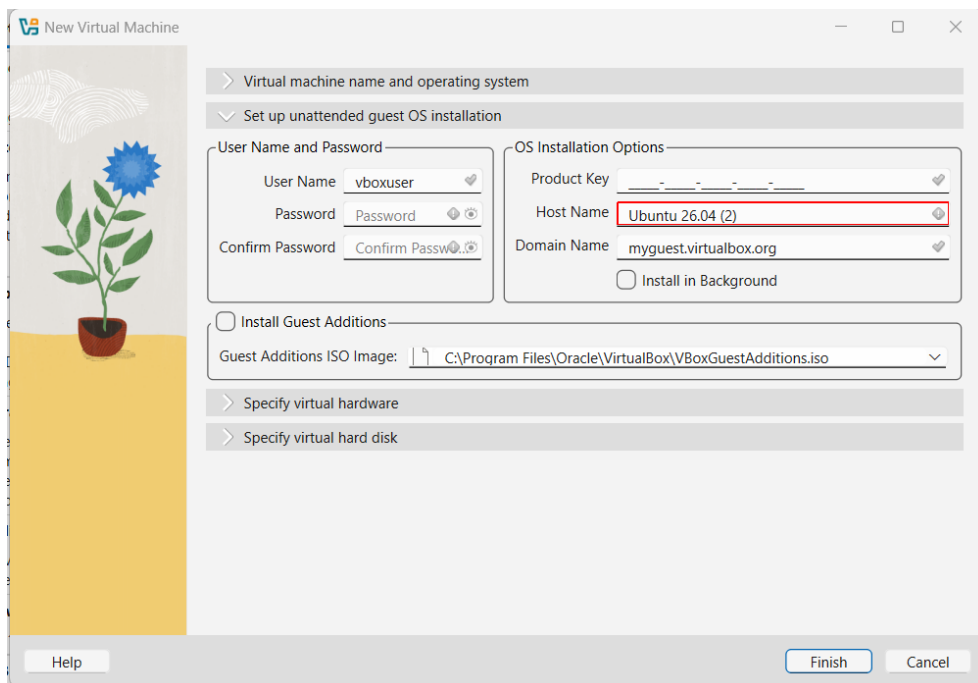
- Create/prepare two Ubuntu Virtual Machines in VirtualBox.
- Configure both VMs on the same virtual network.
- Assign/check the IP address of both VMs.
- Test connectivity using the ping command.
- Designate one VM as the Server.
- Designate the other VM as the Client.
- Create a TCP server program on the Server VM.
- Create a TCP client program on the Client VM.
- Establish a connection between the two VMs.
- Send and receive messages through the chat application.
- Demonstrate the working chat application.
- Capture screenshots of configuration, connectivity and chat communication.

- 1) VM-1 is already created.
- 2) Now I create VM-2.
- 3) Steps of creating VM-2:-

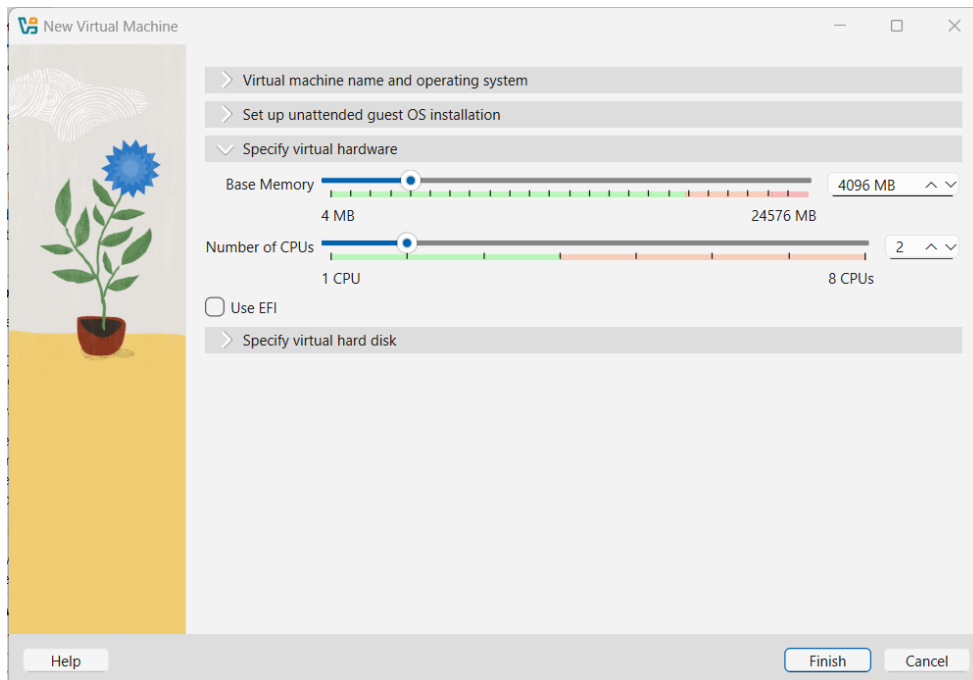
- a) Give VM Name
- b) Give the ISO file



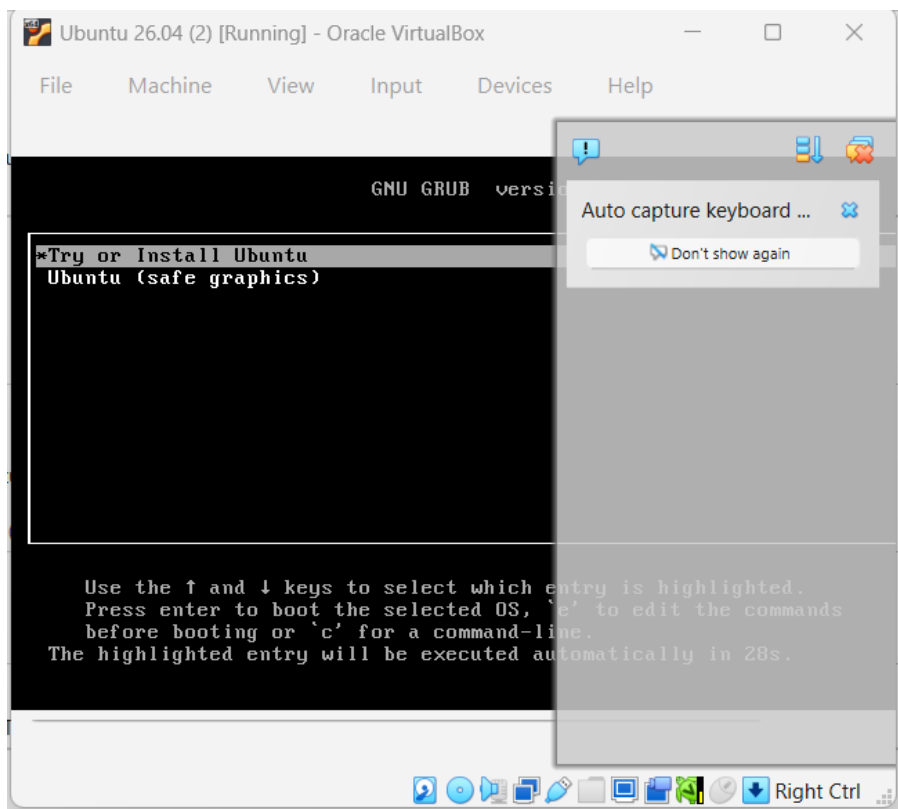
- c) Don't do anything just let it by default



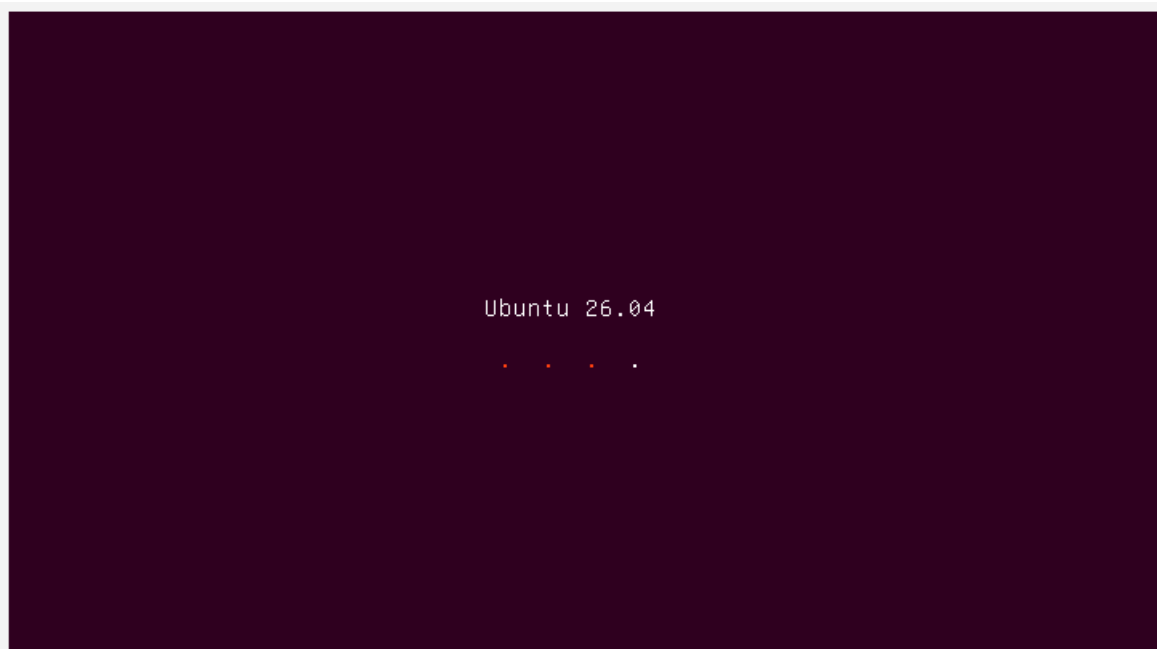
- d) Set the memory = 4 GB or 4096 MB
- e) Set the number of CPUs = 2

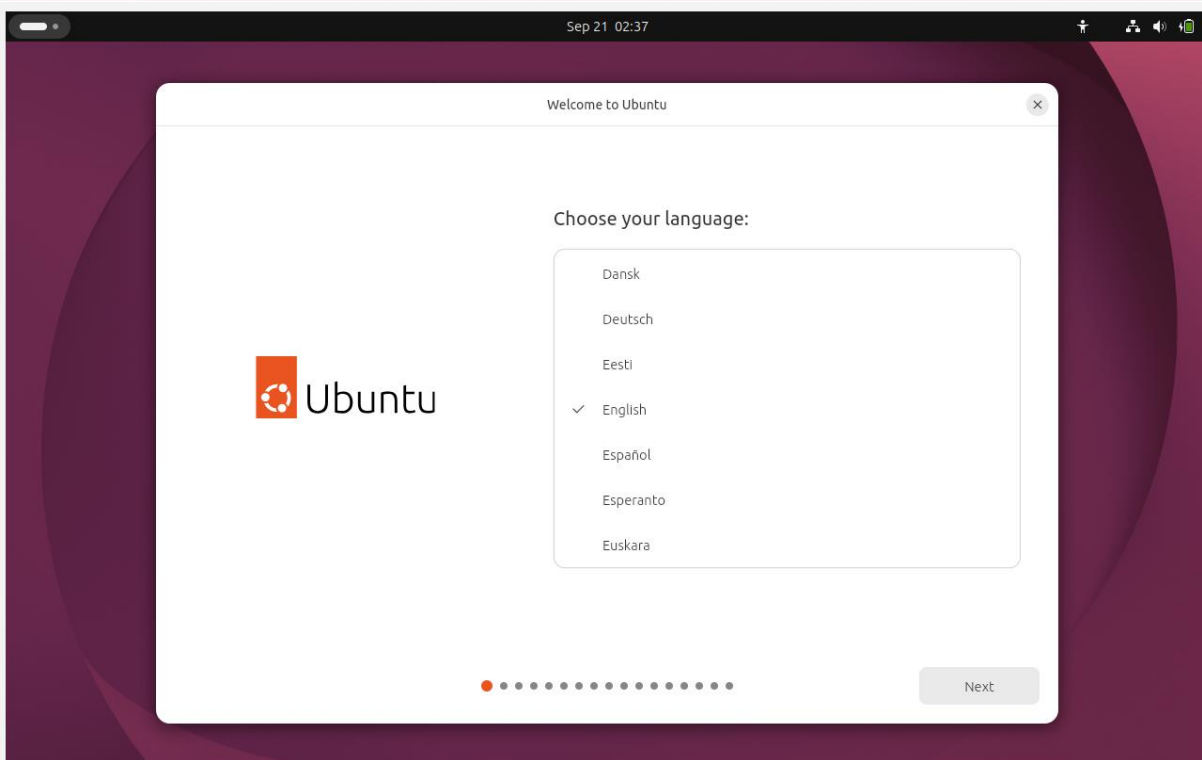
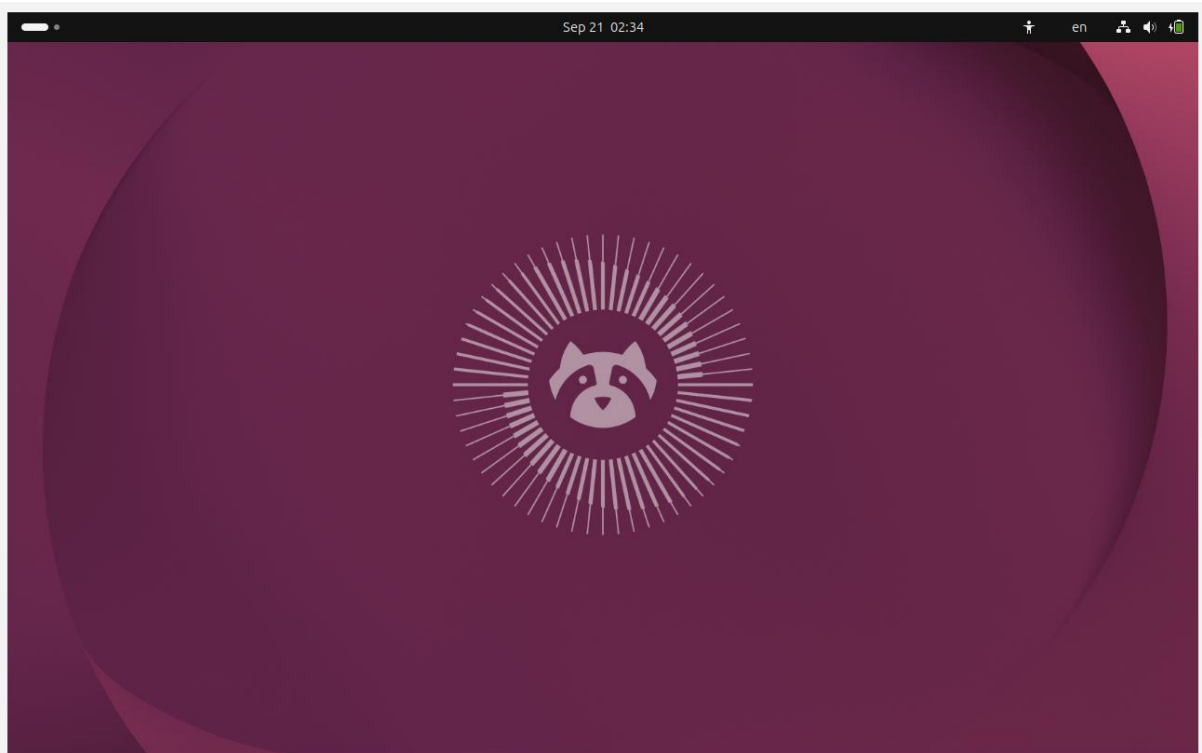


- f) Now Do click on “Try or instal Ubuntu”

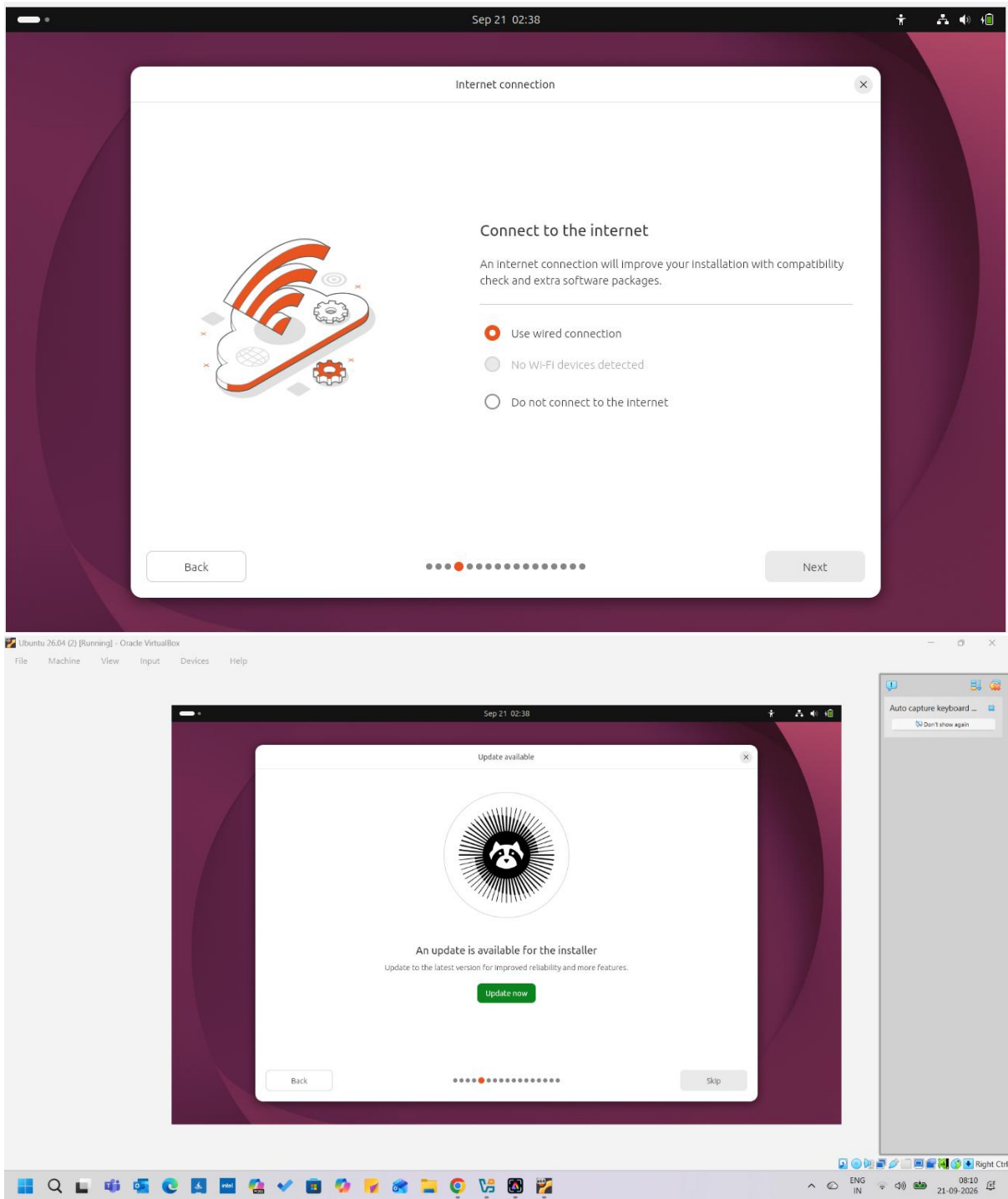


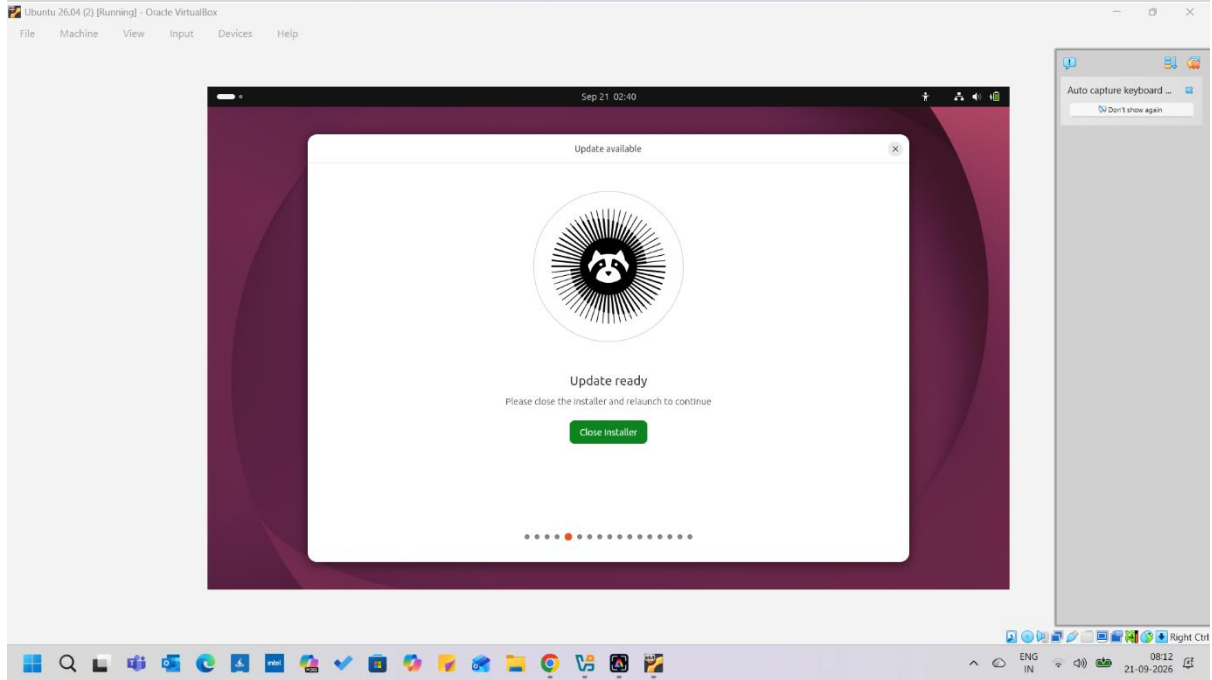
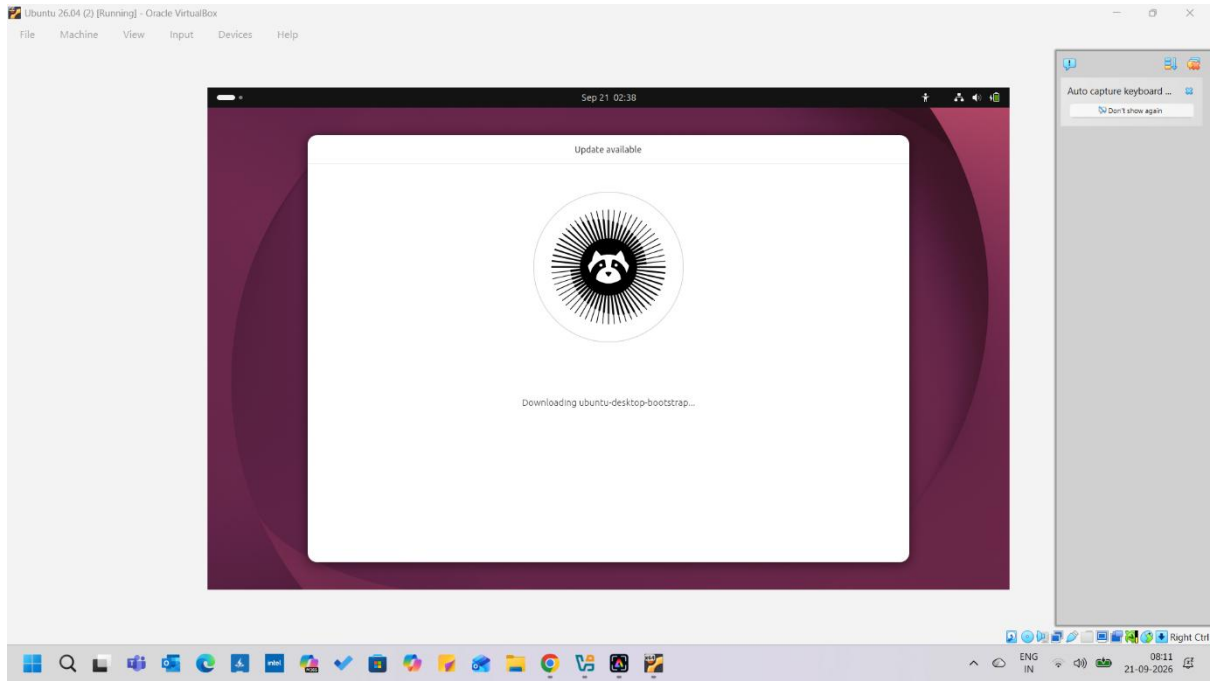
g) Now Ubuntu is Installing

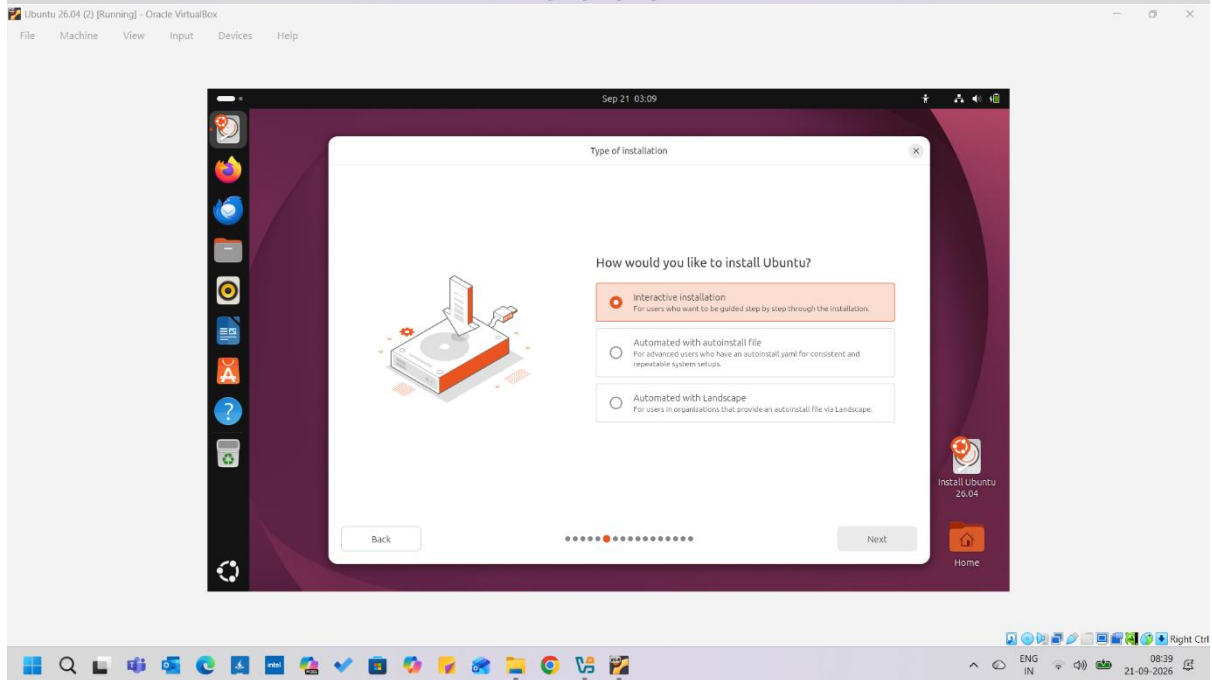
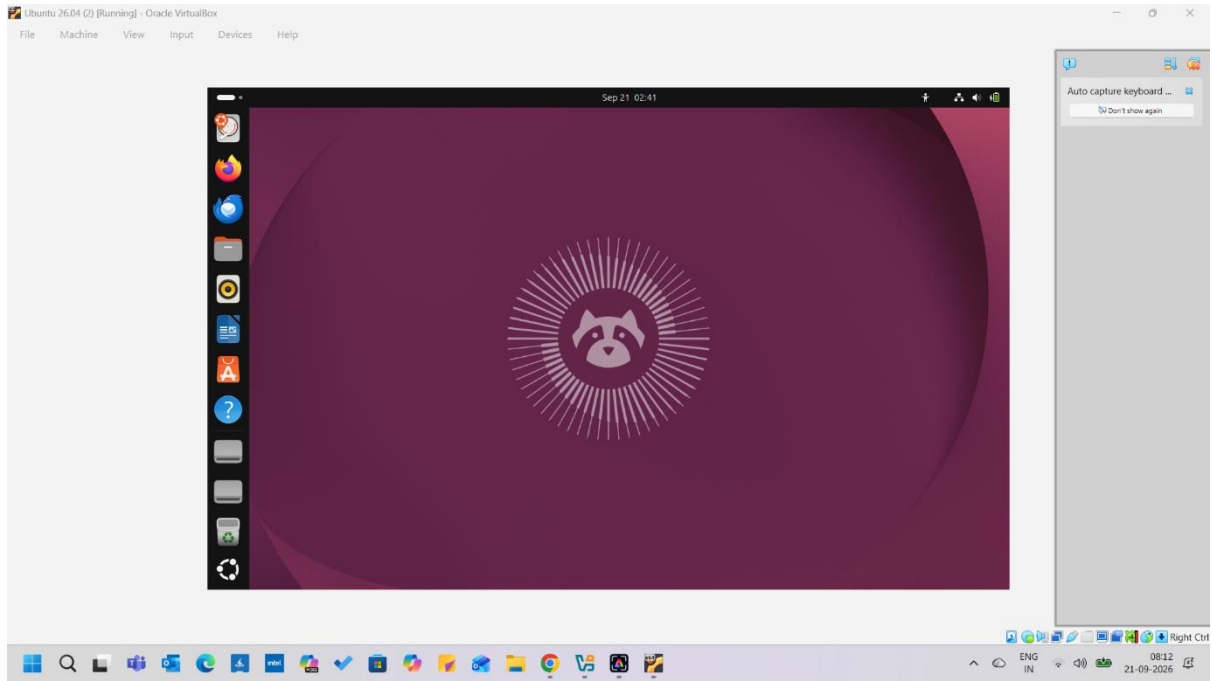


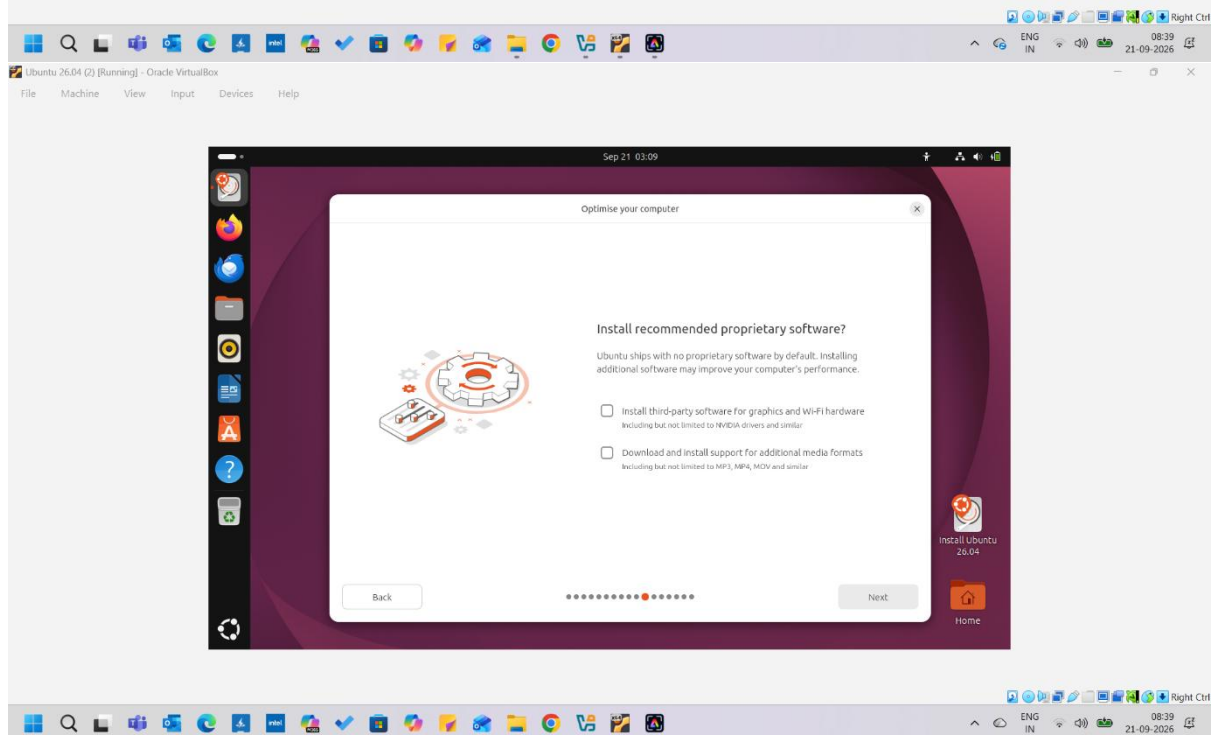
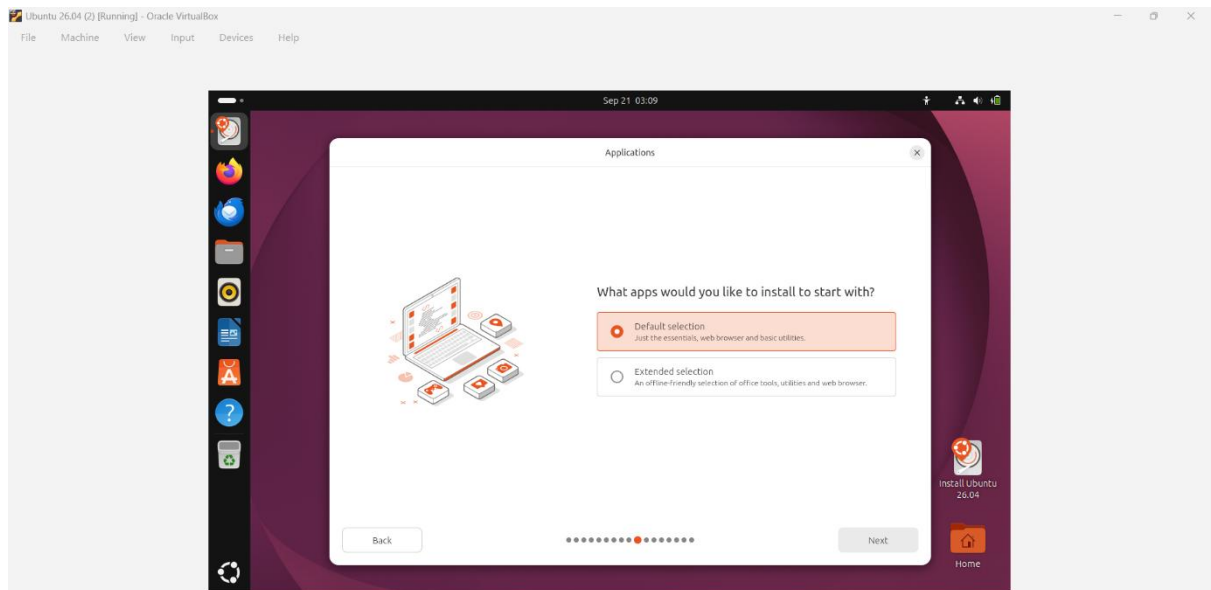


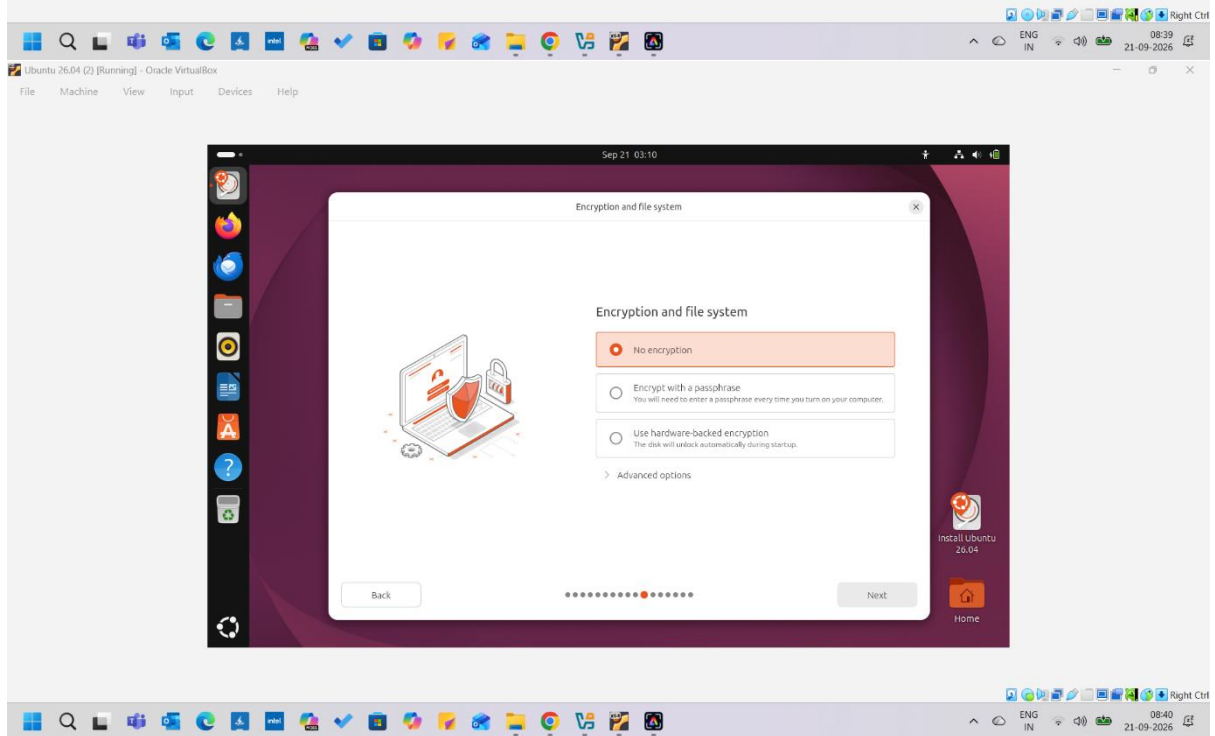
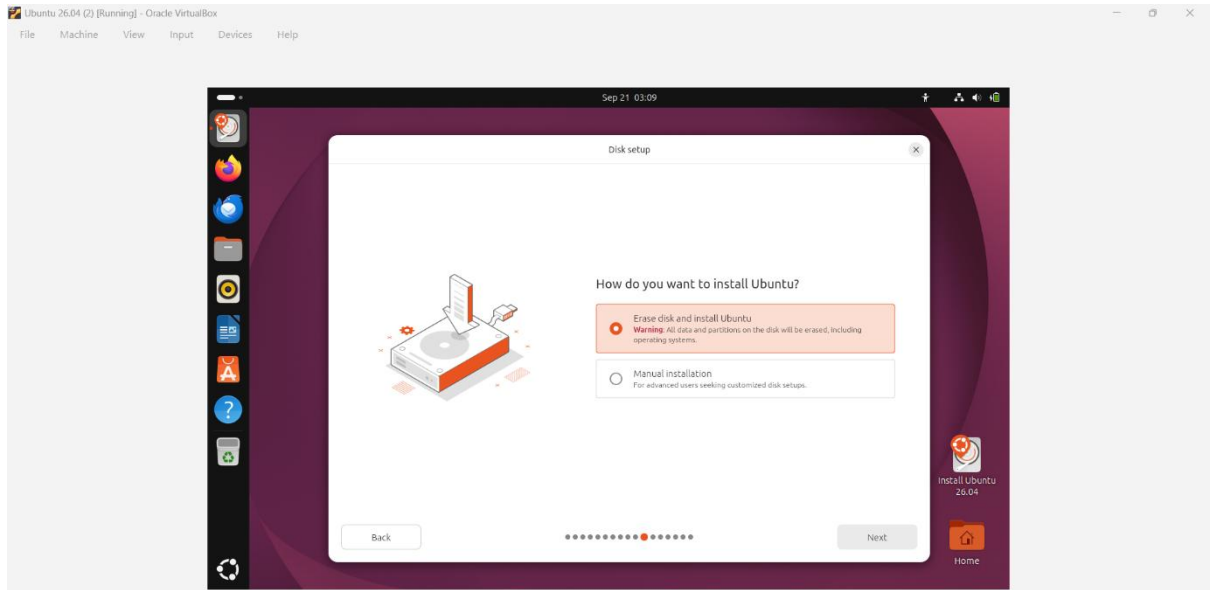


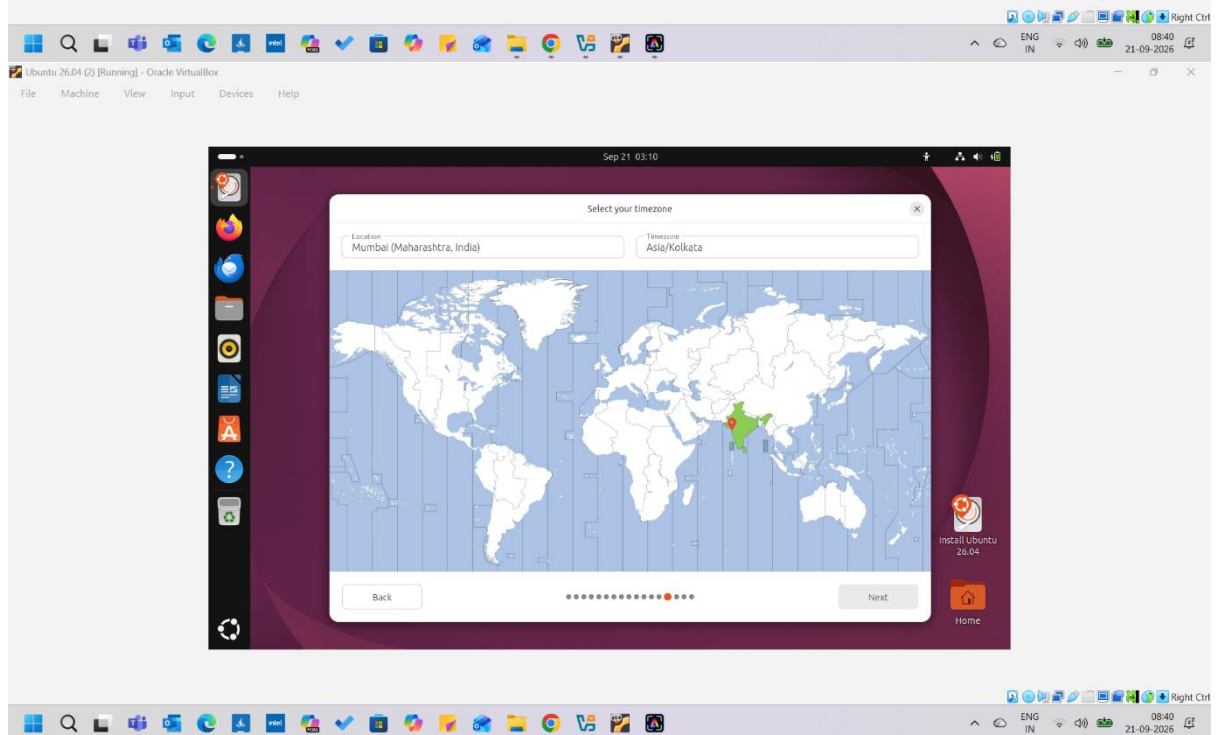
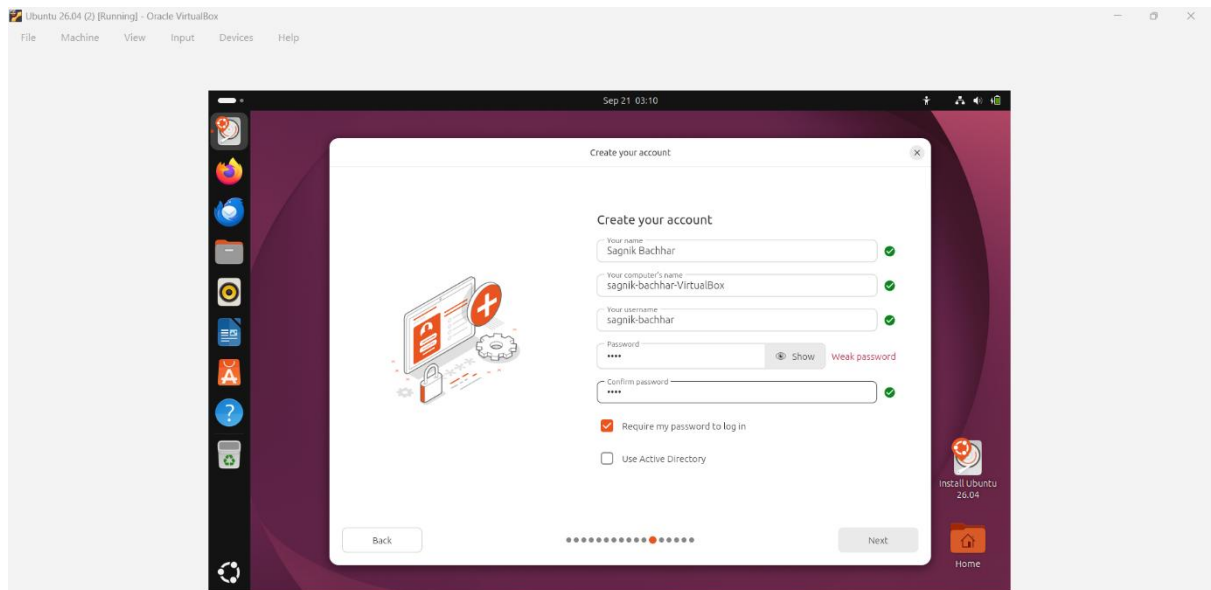


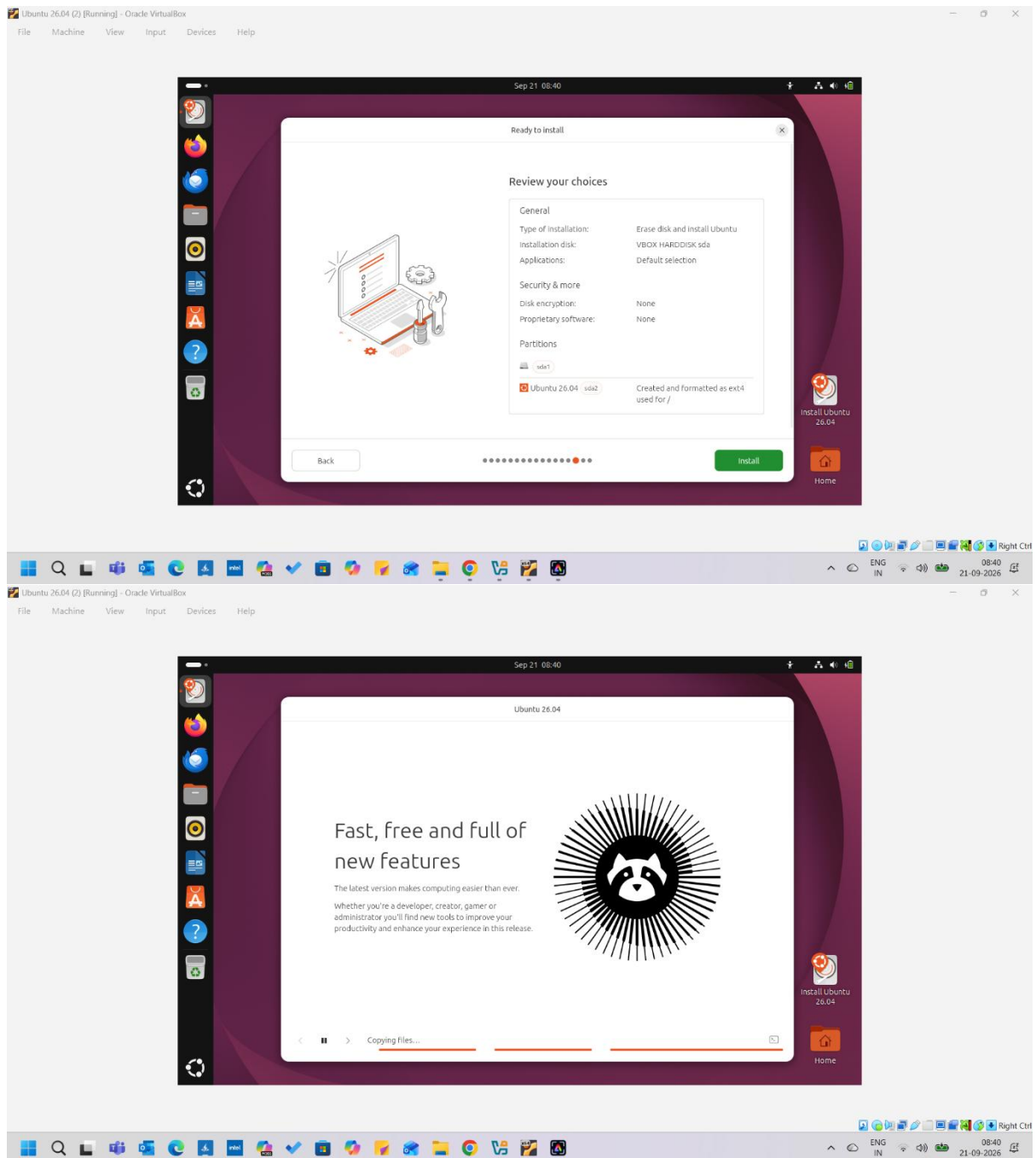






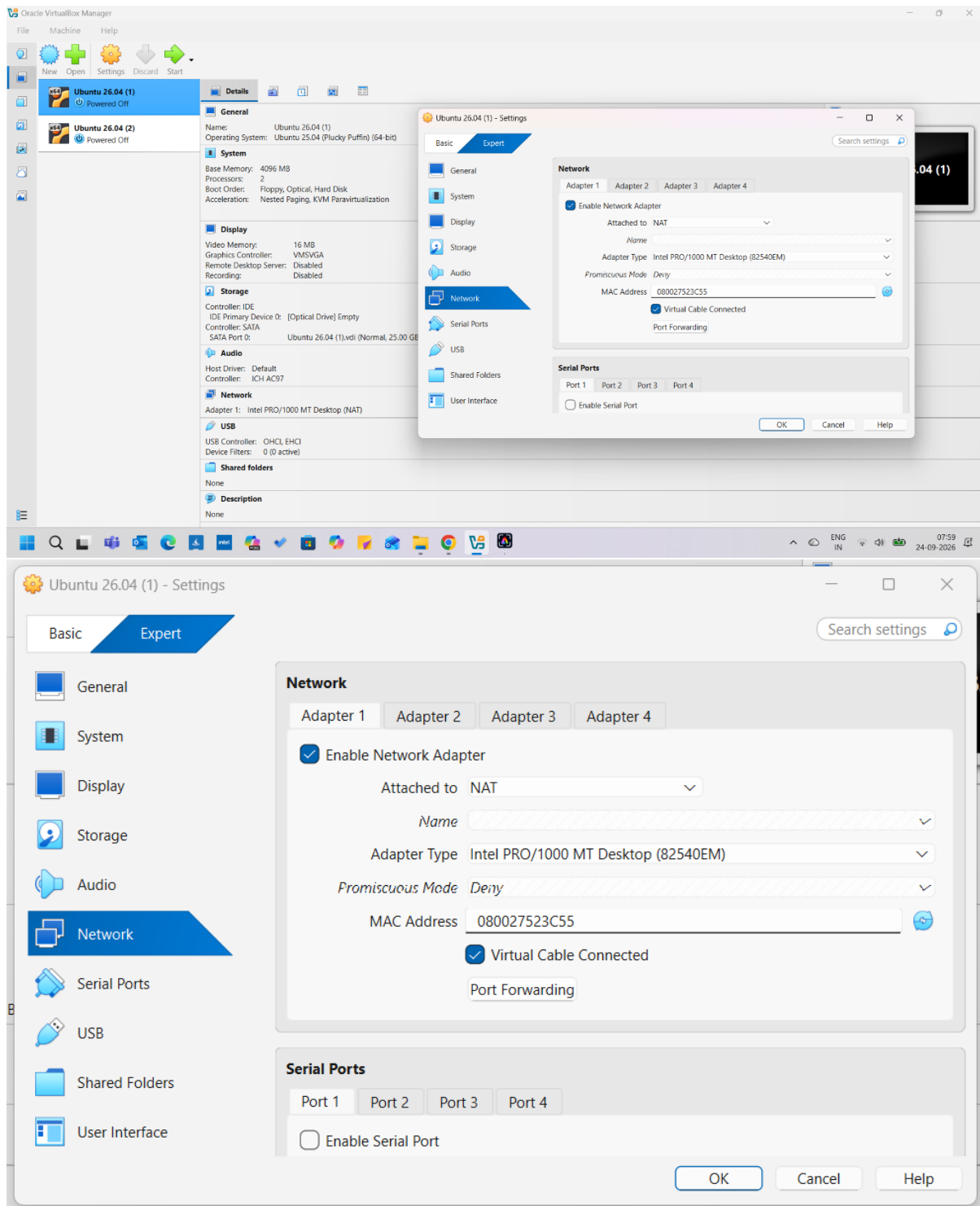


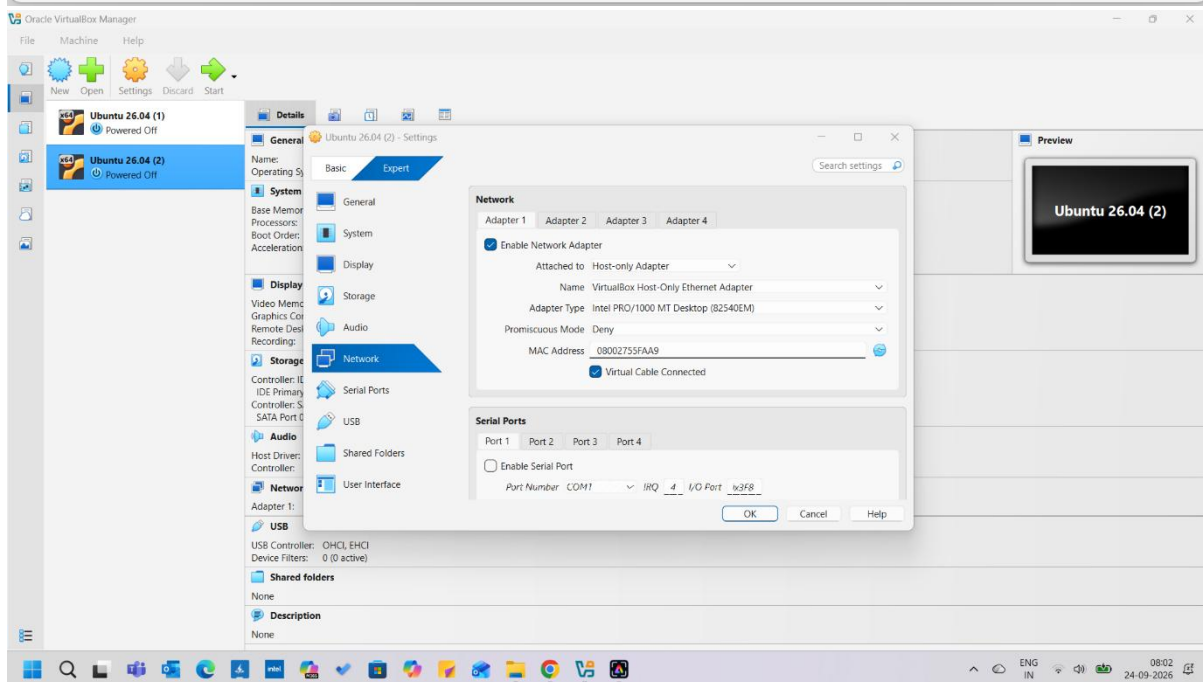
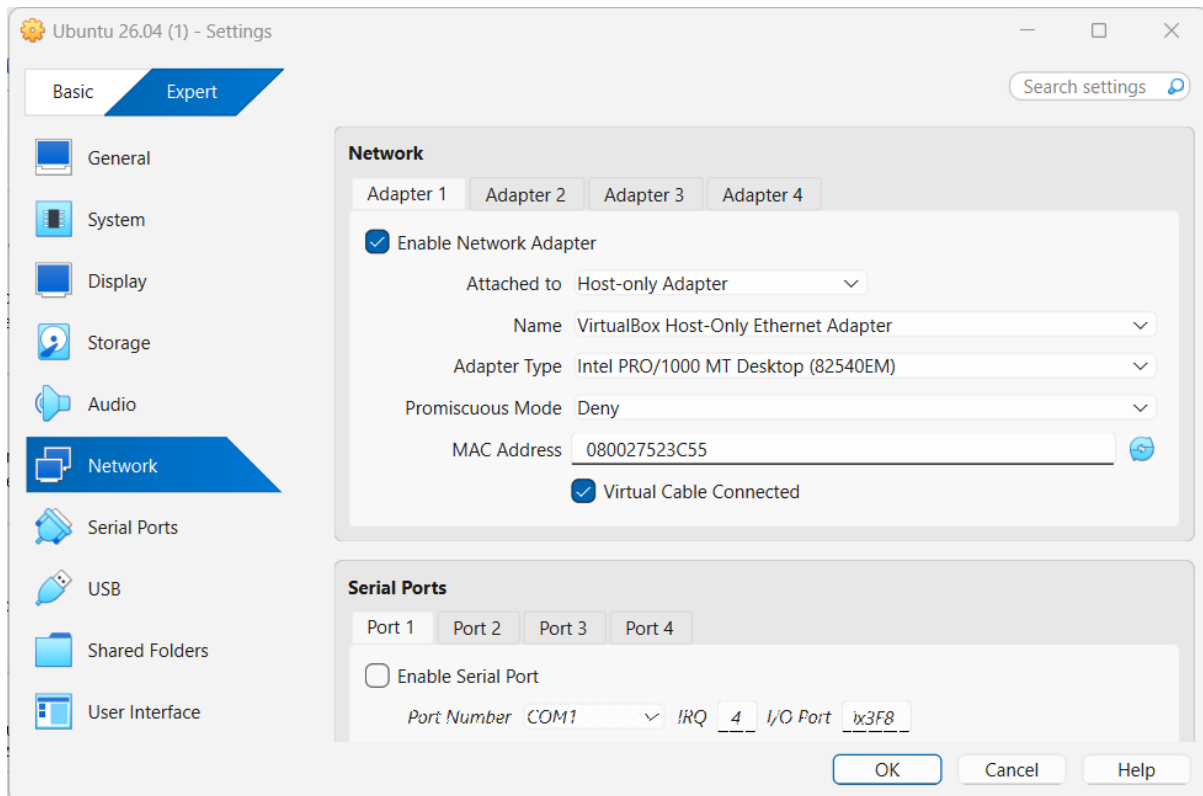


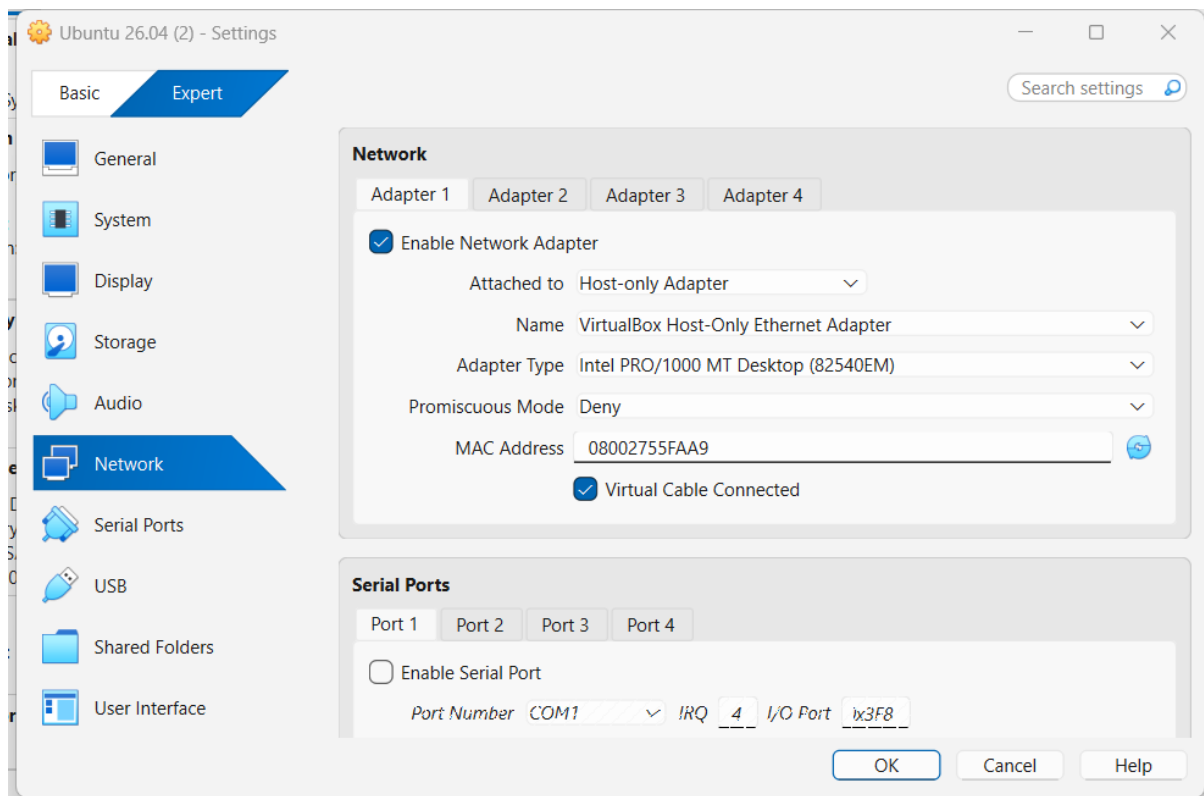


Installation of VM2 Is done.

Now configure Both the VMs on the Virtual network

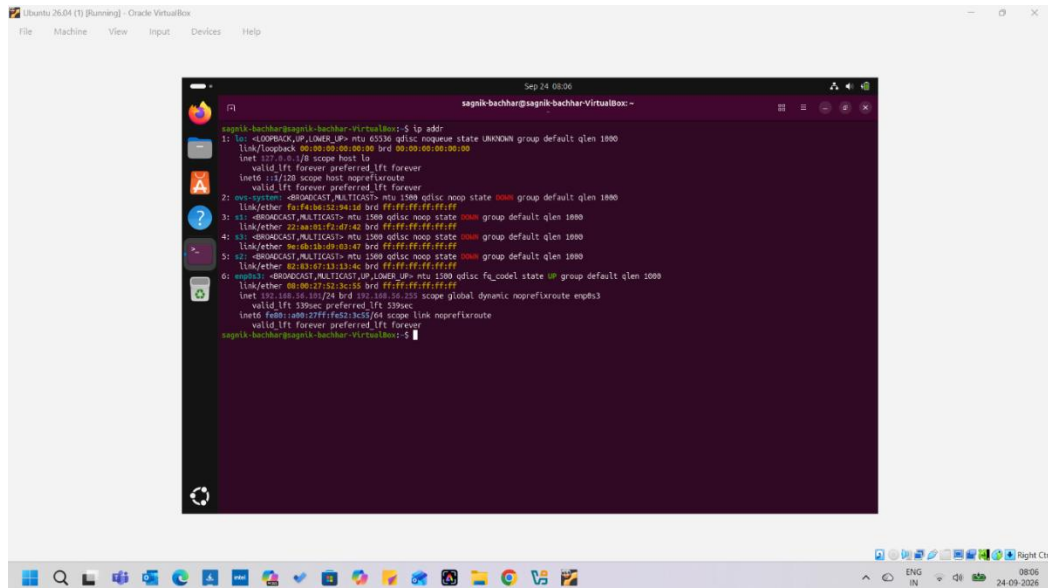






Now Checking IP address Of both the VMs :

VM – 1:



```
Sep 24 08:06
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
   link/ether fa:f4:b6:52:94:1d brd ff:ff:ff:ff:ff:ff
3: s1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
   link/ether 22:aa:01:f2:d7:42 brd ff:ff:ff:ff:ff:ff
4: s3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
   link/ether 9e:6b:1b:d9:03:47 brd ff:ff:ff:ff:ff:ff
5: s2: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
   link/ether 82:83:67:13:13:4c brd ff:ff:ff:ff:ff:ff
6: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:52:3c:55 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
       valid_lft 539sec preferred_lft 539sec
   inet6 fe80::a00:27ff:fe52:3c55/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

## VM – 2 :

```
Ubuntu 26.04 (2) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Sep 24 08:09
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:55:fa:a9 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
       valid_lft 547sec preferred_lft 547sec
   inet6 fe80::a00:27ff:fe55:faa9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```
Sep 24 08:09
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:fa:a9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 547sec preferred_lft 547sec
    inet6 fe80::a00:27ff:fe55:faa9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

Now Test Connectivity using ping command:

VM – 1 :

```
Ubuntu 26.04 (1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Sep 24 08:11
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.51 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.88 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.95 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.30 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.937 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=1.95 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.35 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=2.59 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=2.00 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.30 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=1.28 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=2.07 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=1.10 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.41 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=2.04 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.76 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=1.06 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.988 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.965 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.00 ms
^C
--- 192.168.56.102 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24038ms
rtt min/avg/max/ndev = 0.937/1.691/2.883/0.622 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```
Sep 24 08:11
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.51 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.88 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.95 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.30 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.937 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=1.95 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.35 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=2.59 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=2.00 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.30 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=1.28 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=2.07 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=1.10 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.41 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=2.04 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.76 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=1.06 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.988 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.965 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.00 ms
^C
--- 192.168.56.102 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24038ms
rtt min/avg/max/mdev = 0.937/1.691/2.883/0.622 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

## VM – 2 :

```
Ubuntu 26.04 (2) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Sep 24 08:12
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:55:fa:a9 brd ff:ff:ff:ff:ff:ff
inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
    valid_lft 547sec preferred_lft 547sec
inet6 fe80::a00:27ff:fe55:faa9/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.78 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=2.38 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.06 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.98 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=2.80 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=4.26 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.28 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=1.55 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.982 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=2.17 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=1.73 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.61 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=2.31 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=1.82 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=1.17 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=3.16 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=2.88 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=3.29 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=1.96 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=2.11 ms
^C
--- 192.168.56.101 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19950ms
rtt min/avg/max/mdev = 0.982/2.313/4.278/0.873 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

```
Sep 24 08:12
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:fa:a9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 547sec preferred_lft 547sec
    inet6 fe80::a00:27ff:fe55:faa9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
 64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.78 ms
 64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=2.38 ms
 64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.06 ms
 64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.98 ms
 64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=2.80 ms
 64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=4.26 ms
 64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.28 ms
 64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=1.55 ms
 64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.982 ms
 64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=2.17 ms
 64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=1.73 ms
 64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.61 ms
 64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=2.31 ms
 64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=1.82 ms
 64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=1.17 ms
 64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=3.16 ms
 64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=2.88 ms
 64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=3.29 ms
 64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=1.96 ms
 64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=2.11 ms
^C
--- 192.168.56.101 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19950ms
rtt min/avg/max/mdev = 0.982/2.313/4.278/0.873 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

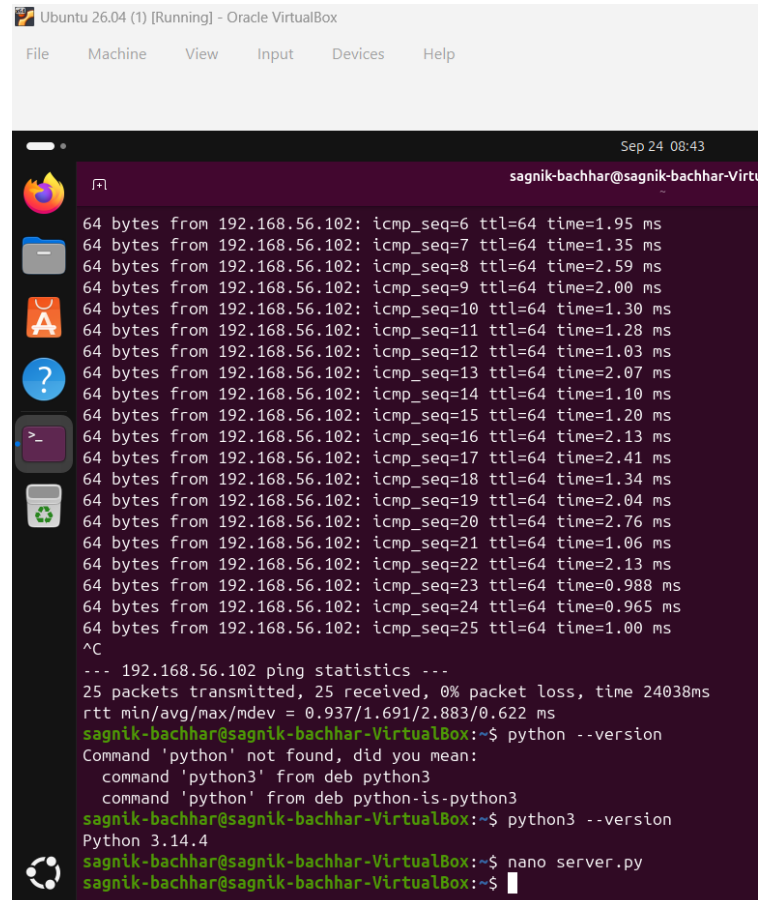
Check the version of python :

```
Ubuntu 26.04 (1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.30 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=1.28 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=2.07 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=1.10 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.41 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=2.04 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.76 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=1.06 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.988 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.965 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.00 ms
^C
--- 192.168.56.102 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24038ms
rtt min/avg/max/mdev = 0.937/1.691/2.883/0.622 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python --version
Command 'python' not found, did you mean:
  command 'python3' from deb python3
  command 'python' from deb python-is-python3
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 --version
Python 3.14.4
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

## Designate VM-1 as Server



Ubuntu 26.04 (1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Sep 24 08:43

sagnik-bachhar@sagnik-bachhar-VirtualBox

```
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=1.95 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.35 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=2.59 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=2.00 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.30 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=1.28 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=2.07 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=1.10 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.41 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=2.04 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.76 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=1.06 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.988 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.965 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.00 ms
^C
--- 192.168.56.102 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24038ms
rtt min/avg/max/mdev = 0.937/1.691/2.883/0.622 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python --version
Command 'python' not found, did you mean:
  command 'python3' from deb python3
  command 'python' from deb python-is-python3
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 --version
Python 3.14.4
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

Write the code for server in VM - 1

Ubuntu 26.04 (1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Sep 24 08:48

sagnik-bachhar@sagnik-bachhar-VirtualBox: ~ — nano server.py

GNU nano 8.7.1 server.py

```
import socket

server_socket = socket.socket(socket.AF_INET , socket.SOCK_STREAM)

server_socket.bind(("0.0.0.0" , 5000))
server_socket.listen(1)

print("Server Started...")
print("Waiting for Client Connection...")

conn, addr = server_socket.accept()
print("Client Connected : " , addr)

while True:
    message = conn.recv(1024).decode()

    if not message:
        break

    print("Client : " , message)

    reply = input("Server : ")
    conn.send(reply.encode())

    if reply.lower() == "exit":
        break
```

[ Wrote 32 lines ]

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^\_ Go To Line M-E Redo M-G Copy

Ubuntu 26.04 (1) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Sep 24 08:48

sagnik-bachhar@sagnik-bachhar-VirtualBox: ~ — nano server.py

GNU nano 8.7.1 server.py

```
server_socket.bind(("0.0.0.0" , 5000))
server_socket.listen(1)

print("Server Started...")
print("Waiting for Client Connection...")

conn , addr = server_socket.accept()
print("Client Connected : " , addr)

while True:
    message = conn.recv(1024).decode()

    if not message:
        break

    print("Client : " , message)

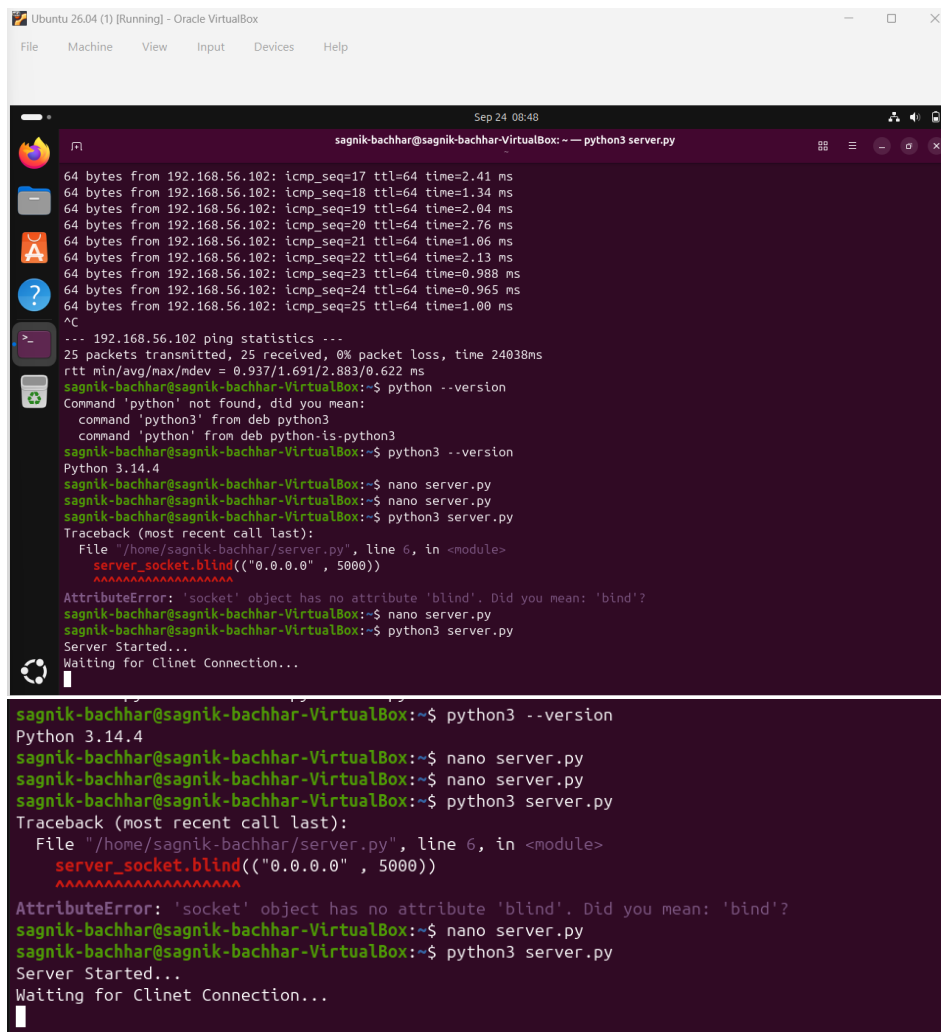
    reply = input("Server : ")
    conn.send(reply.encode())

    if reply.lower() == "exit":
        break

conn.close()
server_socket.close()
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^\_ Go To Line M-E Redo M-G Copy

## Send Connexion request from VM - 1



The screenshot shows a terminal window titled "sagnik-bachhar@sagnik-bachhar-VirtualBox: ~ -- python3 server.py". The terminal output includes:

```
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.41 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=2.04 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.76 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=1.06 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.988 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.965 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.00 ms
^C
--- 192.168.56.102 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24038ms
rtt min/avg/max/mdev = 0.937/1.691/2.883/0.622 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python --version
Command 'python' not found, did you mean:
  command 'python3' from deb python3
  command 'python' from deb python-is-python3
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 --version
Python 3.14.4
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 server.py
Traceback (most recent call last):
  File "/home/sagnik-bachhar/server.py", line 6, in <module>
    server_socket.blind(("0.0.0.0" , 5000))
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AttributeError: 'socket' object has no attribute 'blind'. Did you mean: 'bind'?
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 server.py
Server Started...
Waiting for Clienet Connection...
```

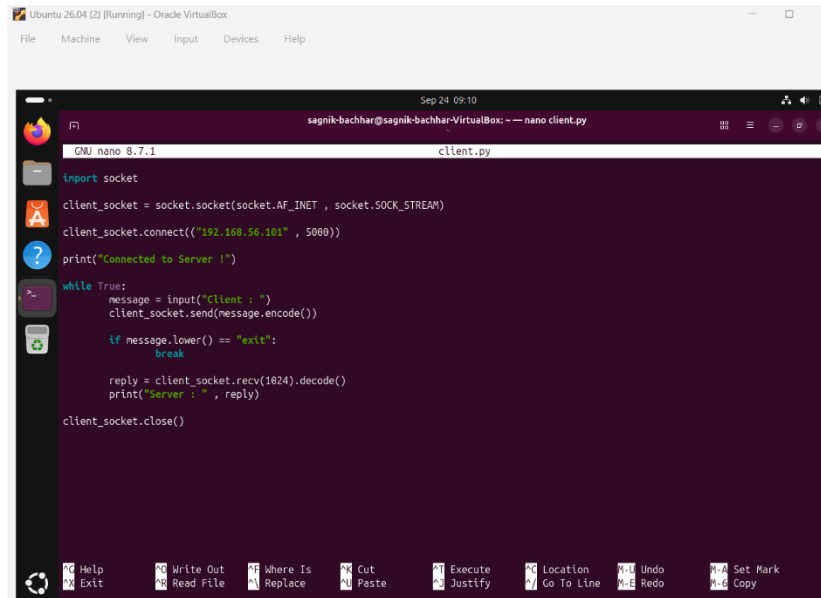
The second terminal window shows the same sequence of commands and the same error message:

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 --version
Python 3.14.4
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 server.py
Traceback (most recent call last):
  File "/home/sagnik-bachhar/server.py", line 6, in <module>
    server_socket.blind(("0.0.0.0" , 5000))
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AttributeError: 'socket' object has no attribute 'blind'. Did you mean: 'bind'?
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 server.py
Server Started...
Waiting for Clienet Connection...
```

## Designate VM-2 as Client

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano client.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 client.py
```

## Write the code for Client in VM – 2



```
GNU nano 8.7.1 client.py
import socket

client_socket = socket.socket(socket.AF_INET , socket.SOCK_STREAM)
client_socket.connect(("192.168.56.101" , 5000))
print("Connected to Server !")

while True:
    message = input("Client : ")
    client_socket.send(message.encode())

    if message.lower() == "exit":
        break

    reply = client_socket.recv(1024).decode()
    print("Server : " , reply)

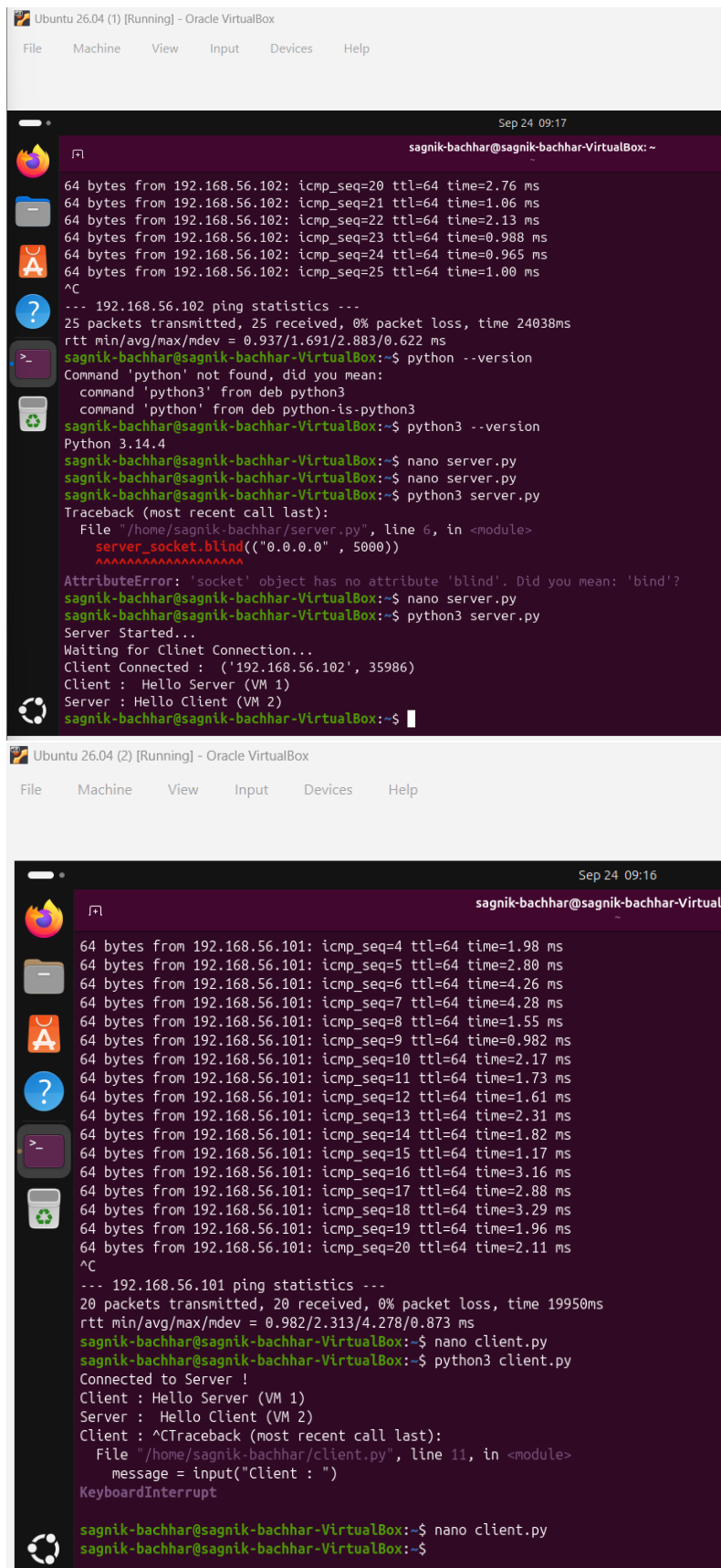
client_socket.close()
```

## Establish The connect in Between VM-1 and VM-2

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano client.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 client.py
Connected to Server !
Client : Hello Server (VM 1)
Server : Hello Client (VM 2)
Client : ^CTraceback (most recent call last):
  File "/home/sagnik-bachhar/client.py", line 11, in <module>
    message = input("Client : ")
KeyboardInterrupt

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano client.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

Sending message from Server to the client & from client to the server :



The image shows two terminal windows from an Oracle VM VirtualBox running Ubuntu 26.04 (1) and (2). The first terminal window shows a series of ping tests from 192.168.56.102 to 192.168.56.101, followed by a ping statistics summary. It then shows the user attempting to run 'python' and 'python3' commands, which fail due to missing or incorrect paths. The user then creates a file named 'server.py' using 'nano' and runs it with 'python3'. The script starts a server on port 5000 and receives a connection from 192.168.56.102. The second terminal window shows a series of ping tests from 192.168.56.101 to 192.168.56.102, followed by a ping statistics summary. It then shows the user creating a file named 'client.py' using 'nano' and running it with 'python3'. The script connects to the server and sends a message. Both windows show the successful execution of the chat application.

```
Ubuntu 26.04 (1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.76 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=1.06 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=2.13 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.988 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.965 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=1.00 ms
^C
--- 192.168.56.102 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24038ms
rtt min/avg/max/mdev = 0.937/1.691/2.883/0.622 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python --version
Command 'python' not found, did you mean:
  command 'python3' from deb python3
  command 'python' from deb python-is-python3
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 --version
Python 3.14.4
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 server.py
Traceback (most recent call last):
  File "/home/sagnik-bachhar/server.py", line 6, in <module>
    server_socket.blind(("0.0.0.0", 5000))
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AttributeError: 'socket' object has no attribute 'blind'. Did you mean: 'bind'?
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano server.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 server.py
Server Started...
Waiting for Client Connection...
Client Connected : ('192.168.56.102', 35986)
Client : Hello Server (VM 1)
Server : Hello Client (VM 2)
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$

Ubuntu 26.04 (2) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.98 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=2.80 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=4.26 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.28 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=1.55 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.982 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=2.17 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=1.73 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.61 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=2.31 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=1.82 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=1.17 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=3.16 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=2.88 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=3.29 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=1.96 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=2.11 ms
^C
--- 192.168.56.101 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19950ms
rtt min/avg/max/mdev = 0.982/2.313/4.278/0.873 ms
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano client.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ python3 client.py
Connected to Server !
Client : Hello Server (VM 1)
Server : Hello Client (VM 2)
Client : ^CTraceback (most recent call last):
  File "/home/sagnik-bachhar/client.py", line 11, in <module>
    message = input("Client : ")
KeyboardInterrupt
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ nano client.py
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

## IP address of Server (VM - 1)

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether fa:f4:b6:52:94:1d brd ff:ff:ff:ff:ff:ff
3: s1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 22:aa:01:f2:d7:42 brd ff:ff:ff:ff:ff:ff
4: s3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 9e:6b:1b:d9:03:47 brd ff:ff:ff:ff:ff:ff
5: s2: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 82:83:67:13:13:4c brd ff:ff:ff:ff:ff:ff
6: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:52:3c:55 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 486sec preferred_lft 486sec
    inet6 fe80::a00:27ff:fe52:3c55/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ hostname -I
192.168.56.101
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```

## IP address of Server (VM - 1)

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:fa:a9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 469sec preferred_lft 469sec
    inet6 fe80::a00:27ff:fe55:faa9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ hostname -I
192.168.56.102
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```